

Lab 4 Activity

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

1

Load in the CO2 data from the `datasets` package. Look at the documentation to understand the variables in this dataset. Identify a way to analyze these data using a t test (several possible choices, but `conc` probably won't be an interesting DV). Evaluate each assumption, conduct the t test, compute an effect size, and interpret the results.

```
data(CO2, package = "datasets")
?CO2
CO2 %>% summary
```

	Plant	Type	Treatment	conc	uptake	
Qn1	: 7	Quebec	:42	nonchilled:42	Min. : 95	Min. : 7.70
Qn2	: 7	Mississippi	:42	chilled :42	1st Qu.: 175	1st Qu.:17.90
Qn3	: 7			Median : 350	Median :28.30	
Qc1	: 7			Mean : 435	Mean :27.21	
Qc3	: 7			3rd Qu.: 675	3rd Qu.:37.12	

```
Qc2      : 7                      Max.      :1000   Max.      :45.50
(Other):42
```

```
# Type or Treatment are appropriate IVs for an ind. samples t-test
# uptake is the most obvious DV
```

```
## Type as the IV:
```

```
# test for outliers:
```

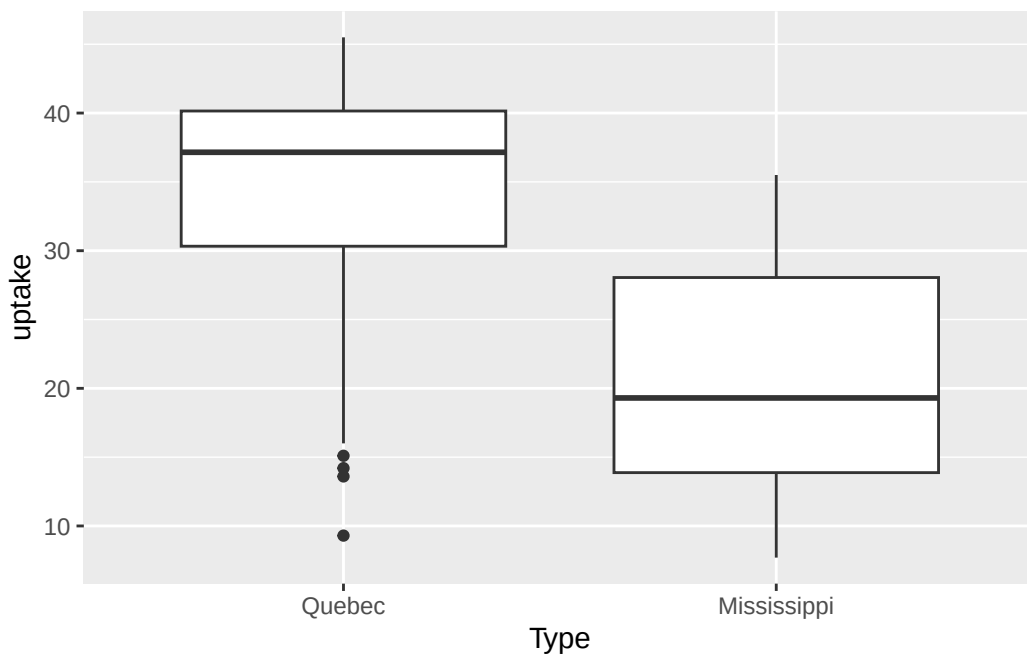
```
C02 %>% group_by(Type) %>%
  summarize(mean = mean(uptake),
            tmean = mean(uptake, trim = .05),
            .groups = "keep") # some differences for Quebec
```

```
# A tibble: 2 x 3
```

```
# Groups:   Type [2]
```

	Type	mean	tmean
	<fct>	<dbl>	<dbl>
1	Quebec	33.5	34.1
2	Mississippi	20.9	20.8

```
ggplot(C02, aes(Type, uptake)) + geom_boxplot()
```



```
# possibly some outliers for Quebec
```

```
# test for normality:
```

```
library(e1071)
```

```
C02 %>% group_by(Type) %>%
```

```
  summarize(mean = mean(uptake), sd = sd(uptake),  
             skew = skewness(uptake), kurt = kurtosis(uptake),  
             n = n(), .groups = "keep")
```

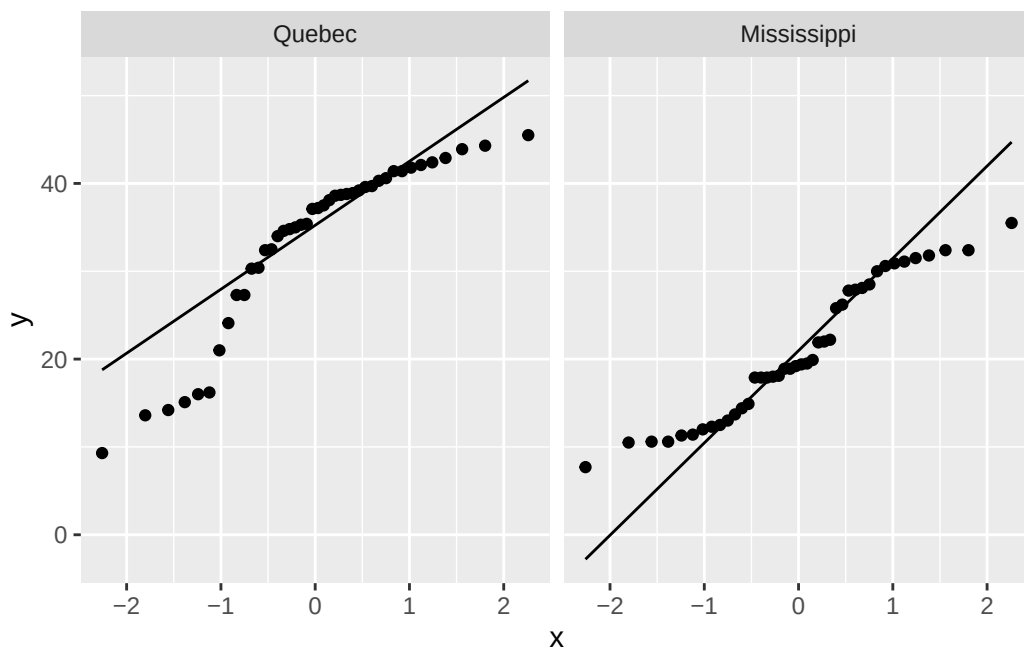
```
# A tibble: 2 x 6
```

```
# Groups:   Type [2]
```

	Type	mean	sd	skew	kurt	n
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	Quebec	33.5	9.67	-1.05	-0.0922	42
2	Mississippi	20.9	7.82	0.172	-1.32	42

```
# some potentially meaningful skew for Quebec, kurt for Mississippi
```

```
ggplot(C02, aes(sample = uptake)) + stat_qq() +  
  stat_qq_line() + facet_wrap(~Type) # some major violations of normality
```



```
# but 42/group might mitigate against this violation

# test for homogeneity of variance:
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

The following object is masked from 'package:purrr':

some

```
leveneTest(uptake ~ Type, CO2)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.1704 0.6808
      82
```

```
# homogeneity not violated, but we'll use Welch anyways

# t-test (two-tailed, Welch)
t.test(uptake ~ Type, CO2)
```

Welch Two Sample t-test

```
data: uptake by Type
t = 6.5969, df = 78.533, p-value = 4.451e-09
alternative hypothesis: true difference in means between group Quebec and group Mississippi
95 percent confidence interval:
 8.839475 16.479572
sample estimates:
mean in group Quebec mean in group Mississippi
      33.54286           20.88333
```

```
# effect size - put pooled standard deviation in the denominator
C02 %>% group_by(Type) %>% summarize(var(uptake), n(), .groups = "keep")
```

```
# A tibble: 2 x 3
# Groups:   Type [2]
  Type      `var(uptake)` `n()`
  <fct>          <dbl> <int>
1 Quebec          93.6    42
2 Mississippi     61.1    42
```

```
sp <- sqrt((41 * 93.6 + 41 * 61.1) / (42 + 42 - 2))
(33.54286 - 20.88333) / sp
```

```
[1] 1.43942
```

```
# mean CO2 uptake rate in Quebec is significantly higher than in Mississippi with a very large effect size
```

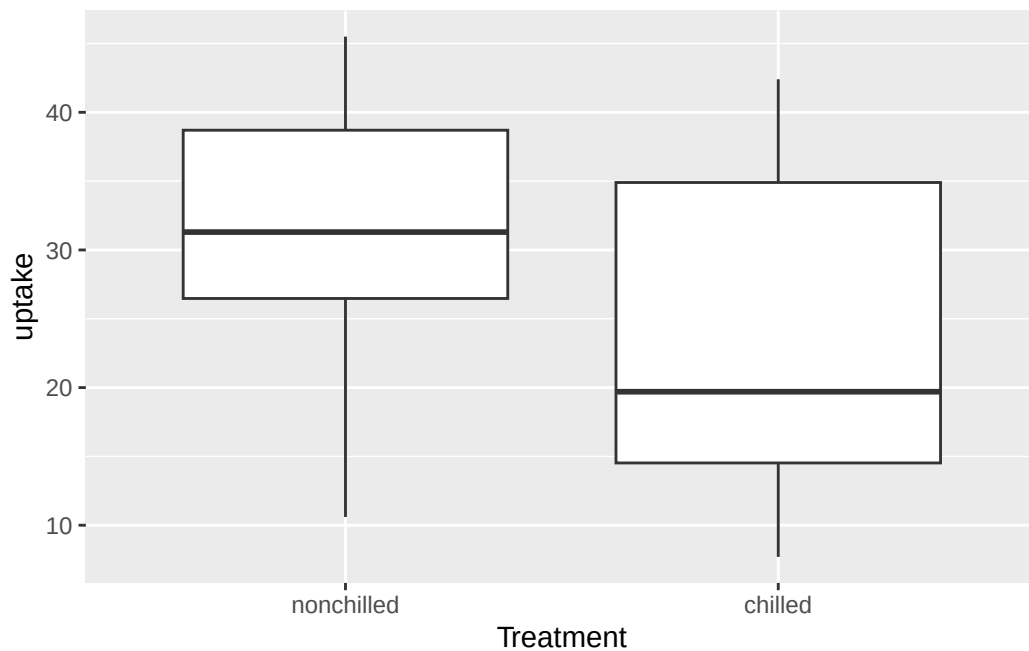
```
## Treatment as the IV:
```

```
# test for outliers:
```

```
C02 %>% group_by(Treatment) %>%
  summarize(mean = mean(uptake),
            tmean = mean(uptake, trim = .05),
            .groups = "keep") # no major differences
```

```
# A tibble: 2 x 3
# Groups:   Treatment [2]
  Treatment    mean tmean
  <fct>      <dbl> <dbl>
1 nonchilled  30.6  30.9
2 chilled    23.8  23.6
```

```
ggplot(C02, aes(Treatment, uptake)) + geom_boxplot()
```

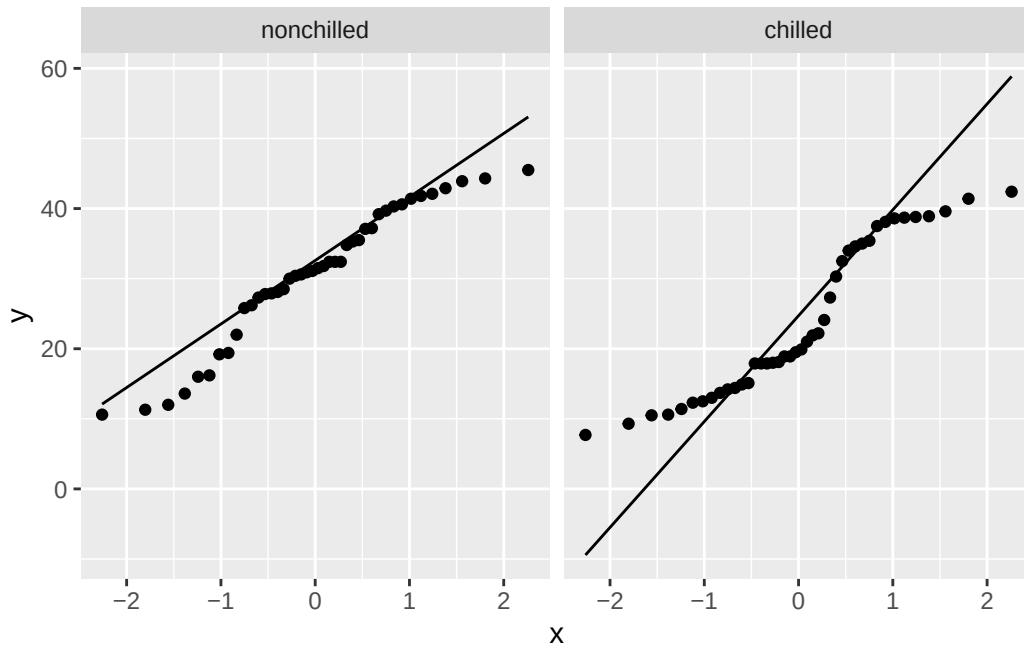


```
# no outliers shown

# test for normality:
library(e1071)
C02 %>% group_by(Treatment) %>%
  summarize(mean = mean(uptake), sd = sd(uptake),
             skew = skewness(uptake), kurt = kurtosis(uptake),
             n = n(), .groups = "keep")

# A tibble: 2 x 6
# Groups:   Treatment [2]
  Treatment mean    sd  skew  kurt    n
  <fct>     <dbl> <dbl> <dbl> <dbl> <int>
1 nonchilled 30.6  9.70 -0.482 -0.706  42
2 chilled   23.8 10.9  0.327 -1.45  42

# some potentially meaningful skew and kurt for both groups
ggplot(C02, aes(sample = uptake)) + stat_qq() +
  stat_qq_line() + facet_wrap(~Treatment) # some major violations of normality
```



```
# but 42/group might mitigate against this violation
```

```
# test for homogeneity of variance:
```

```
library(car)
```

```
leveneTest(uptake ~ Treatment, CO2)
```

```
Levene's Test for Homogeneity of Variance (center = median)
```

```
  Df F value Pr(>F)
```

```
group 1  1.2999 0.2576
```

```
 82
```

```
# homogeneity not violated, but we'll use Welch anyways
```

```
# t-test (two-tailed, Welch)
```

```
t.test(uptake ~ Treatment, CO2)
```

```
Welch Two Sample t-test
```

```
data: uptake by Treatment
```

```
t = 3.0485, df = 80.945, p-value = 0.003107
```

```

alternative hypothesis: true difference in means between group nonchilled and group chilled :
95 percent confidence interval:
  2.382366 11.336682
sample estimates:
mean in group nonchilled    mean in group chilled
          30.64286              23.78333

```

```

# effect size - put pooled standard deviation in the denominator
C02 %>% group_by(Treatment) %>% summarize(var(uptake), n(), .groups = "keep")

```

```

# A tibble: 2 x 3
# Groups:   Treatment [2]
  Treatment `var(uptake)` `n()`
  <fct>          <dbl> <int>
1 nonchilled      94.2    42
2 chilled       118.    42

```

```

sp <- sqrt((41 * 94.2 + 41 * 118) / (42 + 42 - 2))
(30.64286 - 23.78333) / sp

```

```
[1] 0.6659424
```

```

# mean C02 uptake rate for nonchilled treatment is significantly higher than in for the chilled

```

2

Load in the `sleep` data from the `datasets` package. Look at the documentation to understand the variables in this dataset (but try not to peek at the examples section of the documentation). Identify a way to analyze these data using a t test. Evaluate each assumption, conduct the t test, compute an effect size, and interpret the results.

Hint: run the following code to transform the data into “long” format, which will make the dataset easier to work with:

```

sleep2 <- sleep %>%
  pivot_wider(id_cols = ID,
              names_from = group,
              names_prefix = "group",
              values_from = extra)

```



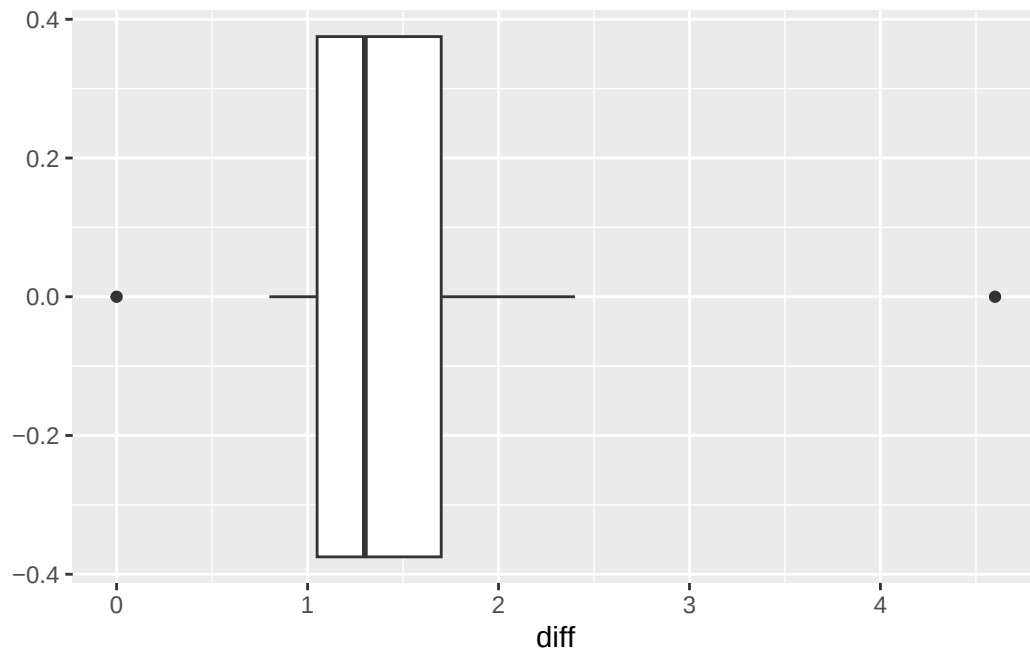
```
data(sleep, package = "datasets")
sleep2 <- sleep %>%
  pivot_wider(id_cols = ID,
              names_from = group,
              names_prefix = "group",
              values_from = extra)
sleep2 %>% summary
```

	ID	group1	group2
1	:1	Min. : -1.600	Min. : -0.100
2	:1	1st Qu.: -0.175	1st Qu.: 0.875
3	:1	Median : 0.350	Median : 1.750
4	:1	Mean : 0.750	Mean : 2.330
5	:1	3rd Qu.: 1.700	3rd Qu.: 4.150
6	:1	Max. : 3.700	Max. : 5.500
(Other):4			

```
# extra should be the DV
# group is an appropriate IV for paired samples t-test (see documentation)

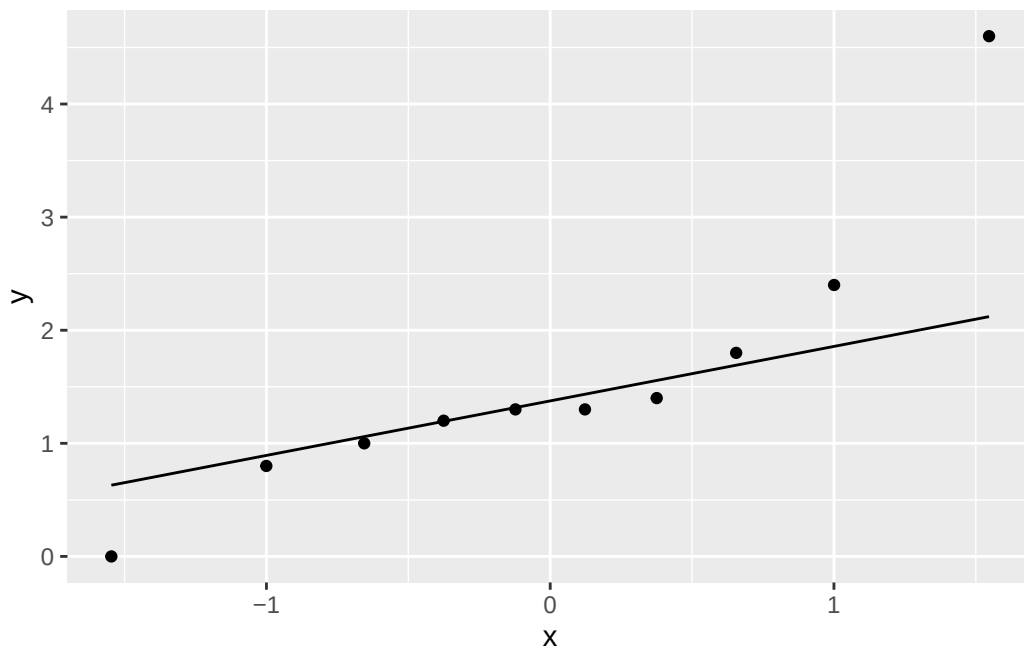
# want to evaluate the normality of difference scores
sleep2 <- sleep2 %>% mutate(diff = group2 - group1)

# trimmed mean may not be appropriate because N is so small
ggplot(sleep2, aes(diff)) + geom_boxplot()
```



possibly likely outliers, but hard to draw any definitive conclusions

```
ggplot(sleep2, aes(sample = diff)) + stat_qq() + stat_qq_line() # probably non-normal
```



```
# an upper-tailed test may be appropriate if we are only concerned about gains
t.test(sleep2$diff, alternative = "greater")
```

One Sample t-test

```
data: sleep2$diff
t = 4.0621, df = 9, p-value = 0.001416
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.8669947      Inf
sample estimates:
mean of x
    1.58
```

```
# effect size
4.0621 / sqrt(10)
```

```
[1] 1.284549
```

```
# there is a significant increase in sleep with drug 2 over drug 1 with a large effect size
```